APPENDIX V

5 pH POINT TITRATION METHOD FOR THE DETERMINATION OF H_2CO_3*ALKALINITY AND SCFA IN AQUEOUS SOLUTIONS CONTAINING KNOWN CONCENTRATIONS OF INORGANIC NITROGEN AND PHOSPHATE

Introduction

The testing procedure for the 5 pH point titration method is set out for the determination of the H_2CO_3*alkalinity (via the total Carbonate species concentration, C_T, and initial sample pH) *and* SCFA in aqueous solutions also containing the ammonium and phosphate weak acid/bases, e.g. anaerobic digester liquid. It was developed for appliCation in monitoring anaerobic fermentation systems. The theory of the method is set out in Chapter 5.

Principle

The sample is titrated from its initial pH to 4 further pH points. Knowing the concentrations of inorganic nitrogen (NH_3/NH_4^+) and phosphate the concentrations of SCFA and H_2CO_3*alkalinity Can be derived from the theory on which the method is based. The measured data are inserted in a personal computer program to facilitate the necessary calculations (see Appendix W).

Apparatus

The following apparatus is required (for details see Appendix B):

- titration burette (10 mℓ) allowing dosing increments of 0,02 ml of titrant,

- pH meter allowing readings to the second decimal place,

- pH probe, preferably of the combined glass electrode type,

- a magnetic stirrer and stirrer bar (length $\simeq$ 25 mm),
- thermometer accurate to ± 0,5 degC,
- filter stand and ordinary filter paper, e.g. Schleicher und Schuell 505,
- measuring pipettes (A grade) from 5 to 50 $m\ell$,
- 100 $m\ell$ Erlenmeyer flask
- specific conductivity meter and probe (if available), and,
- mass scale accurate to ± 1 mg,
- stop watch.

Chemicals

Hydrochloric acid, sodium hydroxide, distilled water, anhydrous sodium carbonate and pH buffer solutions (pH = 4,00 and pH = 7,00).

The hydrochloric acid requires accurate standardisation, see section "Standardization of strong acid". With the sample volumes (for anaerobic digester samples), suggested in the testing procedure below, standardisation of the strong acid to about 0,08 N is recommended, using high purity sodium carbonate. The strong base solution (if required), e.g. sodium hydroxide, should be made up to approximately the same normality as the hydrochloric acid e.g. 0,08 to 0,1 N; standardisation of the strong base is not necessary.

The pH probe needs to be calibrated using two pH buffer solutions bracketing the pH titration range of the 5 pH point titration (excluding the initial pH). It is recommended to use the following two NBS standard buffer solutions, (1) 0,05 M potassium hydrogen phthalate buffer (pH = 4,00 at 25° C), and (2) 0,0275 M disodium hydrogen phosphate and 0,025 M potassium dihydrogen phosphate (pH 7,00 at 25° C).

<u>Sample preparation</u>

A choice has to be made with regard to:

(1) Sample size (ml) undiluted.

(2) Sample size (ml) diluted.

Sample size (undiluted): The undiluted sample has to be representative of the liquid to be tested. With an anaerobic reactor liquid usually a sufficiently large liquid volume is available for testing purpose. Filter the sample prior to the titration to (1) separate the solid from the liquid phase and (2) expel CO_2 from the sample (this has the important effect in that it raises the pH of the undiluted sample, a matter of particular importance in the event that the pH of the undiluted sample is below 6,7). From the filtered undiluted sample a volume needs to be selected for titration.

The choice of undiluted sample volume is determined by (1) the dilution ratio required and (2) the diluted sample volume [see section "Sample size (diluted)" below]. Dilution of the sample is necessary for two reasons: (1) to achieve a total carbonate species concentration, C_T, below about 500 mg/ℓ as $CaCO_3$ to avoid excessive CO_2 loss during titration, (2) to reduce the total dissolved solids concentration to < 2500 mg/ℓ to allow calculation of the activity coefficients according to the Davies Equation (see Appendix A) and, (3) to reduce the temperature of the diluted sample to about 20 – 25°C. For anaerobic reactor liquids, generally the above conditions will be satisfied using a dilution ratio of 1:4. Normally a diluted sample volume of 50 mℓ is adequate (see below); this implies an undiluted sample volume of 10 mℓ. The dilution ratio is kept low to avoid multiplication effects of titration errors: C_T, H_2CO_3*alkalinity and SCFA are determined for the *diluted* sample by titration; the respective values for the undiluted sample are obtained by multiplying the values of the diluted sample with the dilution ratio. Hence any titration error in the diluted sample will be multiplied by the dilution ratio leading to loss of accuracy and precision.

Sample size (diluted): The diluted sample size depends on the physical properties of the pH probe and the titration vessel. In developing the 5 pH point titration method, the pH's were measured using a combined glass electrode (Radiometer Copenhagen GK2401C). The basic physical requirement to be satisfied is that the tip and porous pin (liquid junction) are immersed in the sample below the liquid surface. With the 100 $m\ell$ Erlenmeyer flask as titration vessel, with the Radiometer probe, the tip and porous pin were well immersed using a diluted sample volume of 50 $m\ell$. With titration vessels greater than 100 $m\ell$, greater diluted sample volumes are required. Using Erlenmeyer flasks as titration vessels, in general, it should be noted that the surface (open to atmosphere) to volume ratio of the diluted sample should be small to minimize CO_2 loss; stirring time should be short and the stirring rate gentle in order to minimize CO_2 loss, yet adequate to achieve homogeneous mixing conditions. From practical experience these conditions appear to be satisfied with 50 $m\ell$ of diluted sample in a 100 $m\ell$ Erlenmeyer flask using a magnetic stirrer bar of 25 mm length rotating at approximately 60 rpm.

5 pH point titration procedure

The 5 pH point titration involves the following step by step procedure:

(1) Pipette 10 $m\ell$ of the filtered sample into a 100 $m\ell$ Erlenmeyer flask and pipette 40 $m\ell$ distilled water to give a dilution of 1:4. The requirement is to have C_T diluted to smaller than about 500 mg/ℓ as $CaCO_3$. (With the dilution ratio of 1:4 this condition usually will be satisfied for anaerobic digestor liquids. If subsequently from the analysis it is found that $C_T >$ 500 mg/ℓ as $CaCO_3$ then the test needs to be repeated at a higher dilution ratio estimated from the initial test results).

(2) Insert thermometer and pH probe, stir gently for 15 seconds (s), take temperature reading, remove thermometer and stop stirring.

(3) Wait a further 45 s and record initial pH reading (this gives the probe a total stabilization period of 60 s at the initial pH).

(4) Switch on stirrer and titrate sample from initial pH to pH_1 (6,7 ± 0,1). When pH_1 has been reached, stir for about 30 s, then switch off stirrer, record volume of titrant added from initial pH to pH_1 and take pH_1

reading about 30 s after termination of stirring. The volume of titrant added to titrate from pH initial to pH_1 is designated V_{x1}.

(5) Repeat step (4) to titrate from pH_1 to pH_2 (5,9 ± 0,1), from pH_2 to pH_3 (5,2 ± 0,1) and from pH_3 to pH_4 (4,3 ± 0,1). The volumes of titrant recorded at each pH point (V_{x2}, V_{x3}, V_{x4}) are the cumulative volumes i.e the volume added from the initial pH point to reach the respective lower pH points.

If the initial pH of the diluted sample is below 6,7 the initial pH is recorded; strong base is added to raise the pH to $pH_1 = 6{,}7 \pm 0{,}1$. The volume of strong base added is ignored (V_{x1} at pH_1 is set to zero when entered in the computer program). Acid titration commences from pH_1, i.e. the cumulative volumes for the strong acid titration from zero to V_{x2}, V_{x3} and V_{x4} are the strong acid volumes added to reach pH_2, pH_3 and pH_4 from the starting pH point, pH_1

If the initial pH of the diluted sample is 6,7 ± 0,1 the initial pH equals pH_1 and $V_{x1} = 0$ (V_{x1} is set to zero when entered into the computer program). The cumulative volumes for the strong acid titration V_{x2}, V_{x3} and V_{x4} are the cumulative strong acid volumes added to reach pH_2, pH_3 and pH_4 from the starting pH point, pH_1

Depending on the initial pH of the diluted sample, two different types of titration data tables will be obtained which can be summarized as follows:

(1) Data table for titration with strong acid only (if initial diluted sample pH > 6,7):

V_x	:	pH_x
0	:	initial diluted sample pH (pH_0) > 6,7
V_{x1}	:	pH_1 (6,70 ± 0,1)
V_{x2}	:	pH_2 (5,90 ± 0,1)
V_{x3}	:	pH_3 (5,20 ± 0,1)
V_{x4}	:	pH_4 (4,30 ± 0,1)

(2) Data table for titration with strong acid after addition of a strong base to raise the initial pH to pH_1 or if initial $pH = pH_1 = 6{,}7 \pm 0{,}1$:

V_x	:	pH_x
0	:	initial diluted sample pH (pH_0)
$V_{x1} = 0$	:	pH_1 (after adding of strong base or if initial pH = 6,70 ± 0,1)
V_{x2}	:	pH_2 (5,90 ± 0,1)
V_{x3}	:	pH_3 (5,20 ± 0,1)
V_{x4}	:	pH_4 (4,30 ± 0,1)

The pH values located within the ± 0,1 limit ideally should be aimed for (and are easily attainable even with little experience). In some cases the pH values pH_1, pH_2, pH_3 and pH_4 might be outside the above mentioned tolerance of ± 0,1 pH units, say due to "over enthusiastic" addition of titrant. However, from experience if the sample is being "over titrated" accidentally it is not necessary to repeat the titration provided the pH value is located within 0,2 pH units from its ideal value. If the sample is being "under titrated", step (4) of the step by step procedure may be repeated to obtain pH_1, pH_2, pH_3 or pH_4 within the acceptable limit of ± 0,1 pH units from its ideal pH value.

Ancillary measurements

Inorganic nitrogen total species concentration

To enhance the accuracy of the H_2CO_3*alkalinity and SCFA estimates the influence of the ammonium weak acid/base subsystem on the 5 pH point titration can be minimized by taking into account its total species concentration, N_T, in the algorithm employed to calculate the H_2CO_3*alkalinity and SCFA. The inorganic nitrogen total species concentration can be measured according to Standard Methods (1989), (for reference see Chapter 9). If no measurement is available an approximate estimate of N_T, or N_T = zero, is entered into the computer program. An assessment of error in measurement or neglect of this subsystem on the estimates of H_2CO_3*alkalinity and SCFA, is presented in Chapter 4.

Inorganic phosphate total species concentration

Analogous to the ammonium weak acid/base subsystem the accuracy of the H_2CO_3*alkalinity and SCFA estimates can be enhanced by taking into account the total species concentration of the phosphate weak acid/base subsystem, P_T, which can be measured according to Standard Methods (1989). If no measurement is available an approximate estimate of P_T, or, P_T = zero is entered into the computer program. An assessment of error in measurement or neglect of this subsystem on the estimates of H_2CO_3*alkalinity and SCFA, is also presented in Chapter 4.

Total dissolved solids, TDS, or specific conductivity, SC, of undiluted sample

To calculate the dissociation constants of the different weak acid/bases (carbonate, SCFA, free and saline ammonia and inorganic phosphate) in the computer program, the total dissolved solids, TDS, or alternatively the specific conductivity, SC, of the *undiluted* sample must be entered and hence needs to be measured (see Appendix A). The measurement of one of these parameters means additional experimental effort which may not be justified seeing that the algorithm employed to determine the SCFA and H_2CO_3*alkalinity is not very sensitive to variations in TDS or SC. Hence approximation of these latter two parameters may be justified in many cases. Accordingly measurement of TDS or SC may not be necessary for each filtered sample but done only daily or weekly, depending on the expected variations of the operating conditions of the reactor. This decision must be taken against the background of the accuracy desired for the determination of SCFA and H_2CO_3*alkalinity.

To illustrate the effect of errors in TDS on the calculation of H_2CO_3*alkalinity and SCFA, consider the following example: Take a typical set of titration data for an anaerobic reactor liquid sample: V_S (diluted) = 50 $m\ell$, V_S (undiluted) = 10 $m\ell$, SCFA = 300 mg/ℓ as HAc, H_2CO_3*alkalinity = 1740 mg/ℓ as $CaCO_3$, TDS = 3340 (mg/ℓ), 21°C (after dilution) and $N_T = P_T = 0$ mg/ℓ; Calculate SCFA and H_2CO_3*alkalinity values using various hypothetical values for TDS ranging from 1000 to 7000 mg/ℓ (in the undiluted sample). The results for these calculations are shown plotted in Fig V.1. The plot indicates that SCFA and H_2CO_3*alkalinity determined with the 5 pH point titration method are not sensitive to changes in TDS (or, equivalently SC).

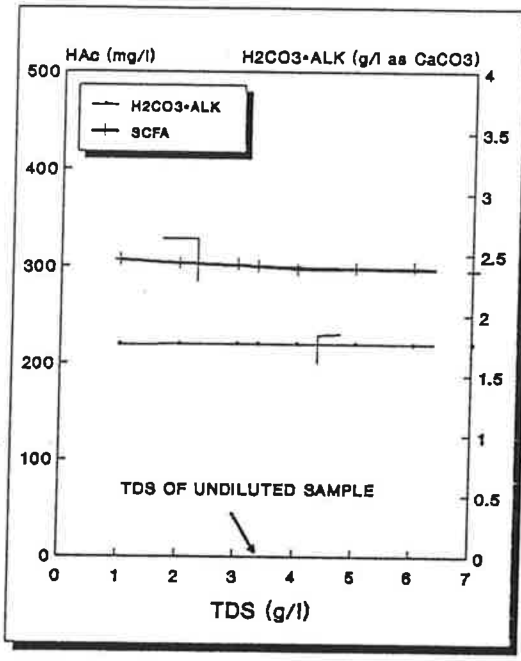

Fig V.1: Influence of changes in the total dissolved solids concentration (TDS) on the determination of SCFA and H_2CO_3*alkalinity with the aid of the 5 pH point titration method: SCFA and H_2CO_3*alkalinity values for a sample containing 300 mg/ℓ of HAc and 1740 mg/ℓ as $CaCO_3$ of H_2CO_3*alkalinity are calculated with hypothetical TDS concentrations ranging from 1000 to 7000 mg/ℓ.

Standardisation of strong acid

To obtain accurate results from the 5 pH point titration method it is of great importance to determine the normality (concentration of H^+ ions) of the strong acid titrant accurately. The standardisation method (used in developing the 5 pH point method) is based on the theory of the H_2CO_3*alkalinity in an aqueous

carbonate solution. The principle of the method is to titrate a solution of known H_2CO_3*alkalinity to the H_2CO_3*alkalinity equivalence point (i.e. the point of zero H_2CO_3*alkalinity) using the strong acid to be standardised. From the mass of strong acid required to titrate to the H_2CO_3*equivalence point, the normality of the strong acid is determined (see below). Using this method one encounters the problem of identifying the H_2CO_3*alkalinity equivalence point.
In Chapter 2 the difficulties in identifying the H_2CO_3*equivalence point were highlighted and it was suggested to use the method developed by Gran (1952), (for reference see Chapter 3) to overcome the problems associated with this endpoint. The theory of Gran's method, more specifically the First Gran Function, was discussed in detail by Loewenthal et al. (1989), (for reference see Chapter 3) and will not be dealt with here; we will deal only with the practical application of the First Gran Function i.e. calculation of the volume of strong acid added to titrate to the H_2CO_3*equivalence point.

The standardisation of the strong acid involves the following steps: (1) preparation of strong acid, (2) Gran titration of carbonate solution, (3) application of the First Gran Function, (4) calculation of normality of strong acid.

Preparation of strong acid: In preparing the strong acid the following two aspects must be kept in mind (1) volume of undiluted sample and (2) the alkalinity of the undiluted sample. Both these factors influence the mass of strong acid required to titrate the sample from the initial pH to pH_4 (in the 5 pH point titration): If the normality is high small quantities of strong acid addition will cause great changes in pH and random errors in strong acid addition will become increasingly significant. If the normality is too low large quantities of strong acid are required to effect pH changes making the titration tedious and mixing of the sample more uncertain. For the purpose of titrating a volume of 10 $m\ell$ of an undiluted sample of a typical anaerobic digester a 0,08 N strong acid is recommended; to approximate this value one may dilute 35 $m\ell$ of 33 percent hydrochloric acid with 5 ℓ of CO_2 free distilled water. This strong acid solution needs to be standardised using the Gran method.

Gran titration of carbonate solution: To standardise the above strong acid, a carbonate solution of known H_2CO_3*alkalinity is made up from CO_2 free distilled water and anhydrous Na_2CO_3: add 5,300 g $NaCO_3$ to 5 ℓ of CO_2 free distilled water; take 50 $m\ell$ of this and perform a Gran titration, as follows: Titrate the

2.12 g Na_2CO_3 to 2 ℓ

sample with the made up strong acid to a pH of about 3,9 and record pH value and the volume of titrant added. Titrate the sample to the final pH of about 3,4 in 4 to 5 titration steps, recording the cumulative volumes of titrant and pH after each step to give a table of titration data, e.g.

V_x	:	pH_x
13,32	:	3,90
13,34	:	3,80
13,37	:	3,70
13,40	:	3,60
13,44	:	3,50

Applying the First Gran Function to this set of data allows the calculation of the volume of strong acid required to titrate to the H_2CO_3*equivalence point.

Application of First Gran Function: Calculate the First Gran Function value, F_{x1}, for each of the V_x, pH_x pairs from,

$$F_{x1} = 10^{-pH_x}(V_s + V_x) \qquad \text{(V.1)}$$

where: F_{x1} = First Gran Function value,
pH_x = pH reading in pH region 3,9 to 3,4,
V_x = volume of strong acid required (mℓ) to titrate to pH_x,
V_s = initial sample volume (mℓ).

This gives the following F_{x1}, V_x pairs:

V_x	:	F_{x1}
13,32	:	0,0080
13,34	:	0,0100
13,37	:	0,0126
13,40	:	0,0159
13,44	:	0,0201

Plot F_{x1} versus V_x; a straight line should be obtained. Draw the best straight line through the data. The interception of the line with the V_x axis, V_{x0}, (i.e. F_{x1} = zero) gives the volume of strong acid required to titrate to the H_2CO_3*equivalence point, i.e. to the point where the H_2CO_3*alkalinity equals zero. (Alternative to the graphical approach V_{x0} can be determined by applying a linear regression to the data set V_x vs F_{x1}; from the linear equation V_{x0} is calculated for $F_{x1} = 0$. Linear regression programs are readily available on many calculators and PC software; in the example presented here this approach was selected and V_{x0} was found to be 13,24 mℓ. With the volume of strong acid known that is required to titrate the carbonate solution of known H_2CO_3*alkalinity, the normality of strong acid can be calculated).

Calculation of normality of strong acid: To calculate the normality of the strong acid note that the mass of H_2CO_3*alkalinity added to the sample equals the mass of strong acid required to titrate to the H_2CO_3*equivalence point:

$$Ca \cdot V_{x0} = m \cdot V_s \qquad \text{(V.2)}$$

where Ca = normality of strong acid (mol/ℓ),

V_{x0} = volume of strong acid required to titrate to H_2CO_3*equivalence point (mℓ),

m = initial H_2CO_3*alkalinity concentration of sample (mol/ℓ),

V_s = initial sample volume (mℓ).

Rearranging Eq (V.2):

$$Ca = m \cdot \frac{V_s}{V_{x0}} \qquad \text{(V.3)}$$

For the above example of 5,300 gNa_2CO_3/(5 ℓ distilled water), initial sample size 50 mℓ and V_{x0} = 13,24 mℓ, Ca can be calculated as follows:

$$Ca = 2 \cdot \frac{\left[\begin{array}{c}\text{mass of } Na_2CO_3 \\ (5{,}3\ g)\end{array}\right]}{\left[\begin{array}{c}\text{Volume of dist.} \\ \text{water } (5\ \ell)\end{array}\right] \left[\begin{array}{c}\text{molecular weight} \\ \text{of } Na_2CO_3\ (106\ g/mol)\end{array}\right]} \cdot \frac{V_s}{V_{x0}}$$

Note that 1 mol of Na_2CO_3 yields 2 moles of H_2CO_3*alkalinity.

1.6g Na_2CO_3 to 5ℓ

0.640g Na_2CO_3 to 2ℓ

Input data table for computer program

The H_2CO_3*alkalinity and SCFA of the undiluted sample is calculated from the titration and ancillary data obtained on the diluted sample (except for TDS and SC which are measured in the undiluted sample); a personal computer program is used for the calculations (The program source code is listed in Appendix W; a floppy disk with the source code and the executable file for the program is attached to the inside cover of the thesis).

The input data for the computer program are as follows:

pH_0 (initial pH of diluted sample)
pH_1 (pH after addition of V_{x1})
pH_2 (pH after addition of V_{x2})
pH_3 (pH after addition of V_{x3})
pH_4 (pH after addition of V_{x4})
V_{x1} (mℓ)
V_{x2} (mℓ)
V_{x3} (mℓ)
V_{x4} (mℓ)
Normality of strong acid (mol/ℓ)
Sample size (undiluted) (mℓ)
Sample size (diluted) (mℓ)
Temperature (deg Celsius)
TDS (mg/ℓ)
SC (mS/m)
N_T (total species con. of free and saline ammonia, mg/ℓ as N)
P_T (total species con. of inorganic phosphate, mg/ℓ as P)

Note that either TDS or SC needs to be entered in the program: if TDS is entered SC is calculated internally and similarly, TDS is calculated if SC is entered.

Output data table of computer program

(1) H_2CO_3*alkalinity of undiluted sample (mg/ℓ as $CaCO_3$)
(2) SCFA of undiluted sample (mg/ℓ as acetic acid)
(3) Correction for systematic pH error

Note: (1) The correction for systematic pH error is subject to the condition that the carbonate weak acid/base dominates in the sample i.e. SCFA (mg/ℓ as acetic acid) needs to be smaller than about 0,5 · C_T (mg/ℓ as $CaCO_3$), see Chapter 5. This condition is tested internally by the program. If the condition is not met no correction for systematic pH error is made by the program and a message is displayed in the output data table that no correction has been made; the resulting loss in accuracy in the remaining output data will however remain within acceptable limits for the purpose of monitoring of anaerobic digesters. (This situation will develop only when the digestor is showing signs of failure, in which event accuracy of SCFA and H_2CO_3*alkalinity no longer is a prime requirement).

The maximum correction for systematic pH errors made by the program is set to ± 0,2 pH units. If the systematic pH error is greater than ± 0,2 pH units this is displayed on the screen with the recommendation that the probe should be recalibrated. From practical experience (with samples from anaerobic digesters) an average systematic pH error of about – 0,05 pH units was encountered. It would appear therefore that with systematic pH errors of ± 0,2 indicated by the computer program, the pH probe most likely is in error and needs to be recalibrated, and the test repeated.

(2) The H_2CO_3*alkalinity and SCFA are the concentrations present in the undiluted sample and these are also the concentrations in the reactor. The concentration of C_T in the undiluted sample, however, is *not* equal to the concentration in the reactor due to CO_2 loss between sampling and testing. If C_T in the *reactor* is to be obtained, this is found from the reactor *in situ* pH, temperature, TDS (or SC) of the reactor liquid and the undiluted H_2CO_3*alkalinity determined with the 5 pH point titration method, see Chapter 2.

<u>Example</u>

The following example demonstrates the data input and data output (after calculation of H_2CO_3*alkalinity, SCFA and the systematic pH error with the aid of the computer program) for data measured on a UASB reactor effluent treating lauter tun (brewery) waste.

Data input screen:

pH_0 (initial pH of diluted sample).....	8,77
pH_1 (pH after addition of V_{x1}).....	6,80
pH_2 (pH after addition of V_{x2}).....	5,92
pH_3 (pH after addition of V_{x3}).....	5,25
pH_4 (pH after addition of V_{x4}).....	4,12
V_{x1} (mℓ).....	1,40
V_{x2} (mℓ).....	3,60
V_{x3} (mℓ).....	4,56
V_{x4} (mℓ).....	4,96
Normality of strong acid (mol/ℓ).....	0,0728
Sample size (undiluted) (mℓ).....	10
Sample size (diluted) (mℓ).....	50
Temperature (deg Celsius).....	24
TDS (mg/ℓ).....	
SC (mS/m).....	380
N_T (total species con. of free and saline ammonia, mg/ℓ as N).....	30
P_T (total species con. of inorganic phosphate, mg/ℓ as P).....	25

Data output screen:

H_2CO_3*alkalinity of undiluted sample
(mg/ℓ as $CaCO_3$)......................................1741

SCFA of undiluted sample
(mg/ℓ as acetic acid)..................................6

systematic pH error..............................– 0,02

APPENDIX W

SOURCE CODE LISTING OF PERSONAL COMPUTER PROGRAM FOR 5 pH POINT TITRATION METHOD

This appendix contains the listing of the source code for the source code file: 5ppt.pas. This file is available on the floppy disk which is attached to the inside of the thesis cover. The program is coded using Turbo Pascal Ver 4.0. It allows the calculation of H_2CO_3*alkalinity, SCFA (as A_T) and systematic pH measurement error from the data collected in the 5 pH point titration procedure. (For 5 pH point titration procedure, see Appendix V).

```
program atct;

                    {5 pH POINT TITRATION PROGRAM}

{This program calculates H2CO3-alkalinity and short-chain fatty acid
 concentrations in aqueous solutions containing the carbonate, SCFA,
 ammonium and phosphate subsystems. The total species concentrations
 of the ammonium and phosphate subsystems need to be entered into the
 computer program. The sample is titrated from its initial pH to four
 further pH values measured in the pH region:  6.7, 5.9, 5.2, 4.3. The
 program provides an estimate of a possible systematic pH measurement
 error in the event that the carbonate system dominates
 over the remaining weak acid/base subsystems.}

{PROGRAM START:}

uses  crt;

type phstr= string[50];

var  pH0, pH1, pH2, pH3, pH4, delph, pHcorr,
     A1, B1, A2, B2,
     MCt1, MCt2, Ct2, Ct1, Ctcomp, MAt1, MAt2, At1, At2, CtAtratio,
     dil, vx1, vx2, vx3, vx4, vxfi, vxs, ca, temp, ktemp, TDS,
     logf1, logf2, pK1, pK11, pK2, pK22, pKn, pKnn, Nt,
     pKp, pKpp, Pt, pKa, pKaa, At,
     MH2CO3alksam, H2CO3alksam, Mcarb01, Mcarb02,
     counter, ionstr, number, vsdil, vsundil :real;

    again, move,intro, x, y, a, b, c, d : char;
    labels: array[1..18] of phstr;
    value: array[1..18] of real;
    I,II,posy: Integer;



function log(x:real):real;
 begin
  log:=ln(x)/2.302585093;
 end;

function tento(y:real):real;
 begin
  tento:=exp(y*ln(10));
 end;

function mue (TDS,dil:real):real;
 begin
  mue:= 0.000025* (TDS/dil-20);
 end;

function logf (mue,ktemp:real):real;
 begin
  logf:= - 1.825 * 1000000 * exp(-1.5*ln(78.3*ktemp)) * (sqrt(mue)/(1+
        sqrt(mue)) - 0.3*mue);
 end;

function perH2CO3 (ph:real):real;
 begin
```

```
  perH2CO3:= 1/(1 + tento(-pk11)/tento(-ph) + tento(-pk11) * tento
              (-pk22)/tento(-2*ph));
 end;

function perHCO3 (ph:real):real;
 begin
  perHCO3:= 1/(tento(-ph)/tento(-pk11) + 1 + tento(-pk22)/tento
           (-ph));
 end;

function perCO3 (ph:real):real;
 begin
  perCO3:=1/(tento(-2*ph)/(tento(-pk11)*tento(-pk22)) + tento
             (-ph)/tento(-pk22) + 1);
 end;

function per(ph,pkk:real):real;
 begin
  per:=tento(-pkk)/(tento(-pkk)+tento(-ph));
 end;

function dH2CO3alk(pHf,pHs: real): real;
 begin
  dH2CO3alk:= perHCO3(pHf)-perHCO3(pHs) + 2 * (perCO3(pHf)-perCO3(pHs));
 end;

function dHAcalk(pHf,pHs: real): real;
 begin
  dHAcalk:= per(pHf,pkaa) - per(pHs,pkaa);
 end;

function MH2O (vxfi,vxs,pHfi,pHs: real):real;
 begin
  MH2O:= (vsdil+vxs)*tento(-pHs)/tento(logf1) - (vsdil+vxfi)*tento(-pHfi)/
        tento(logf1) + (vsdil+vxfi)*tento((pHfi-14)) - (vsdil+vxs)*
        tento((pHs-14));
 end;

function MNH3(pHf,pHs:real):real;
 begin
  MNH3:= Nt/(14000*dil) * vsdil * (per(pHf,pKnn) - per(pHs,pKnn));
 end;

function MHPO4(pHf,pHs:real):real;
 begin
   MHPO4:= Pt/(31000*dil) * vsdil * (per(pHf,pKpp) - per(pHs,pKpp))
 end;



procedure mark;
 begin
  textbackground(white);
  textcolor(black);
 end;

procedure unmark;
 begin
  textbackground(black);
  textcolor(white);
 end;

procedure box(x1,y1,x2,y2,a:integer);
 var
  II:integer;
 begin
  a:= x2-x1-1;
   FOR II:=1 to a DO          {draw the horizontal lines}
```

```
    begin
     GOTOXY(x1+1,y1);
     writeln(chr(196));
     GOTOXY(x1+1,y2);
     writeln(chr(196));
     x1:=x1+1;
    end;
  x1:=x1-a;
  a:=y2-y1-1;
   FOR II:=1 to a DO          {draw the vertical lines}
    begin
     GOTOXY(x1,y1+1);
     writeln(chr(179));
     GOTOXY(x2,y1+1);
     writeln(chr(179));
     y1:=y1+1;
    end;
  y1:=y1-a;
  GOTOXY(x1,y1);              {draw the corners}
  writeln(chr(218));
  GOTOXY(x1,y2);
  writeln(chr(192));
  GOTOXY(x2,y1);
  writeln(chr(191));
  GOTOXY(x2,y2);
  writeln(chr(217));
end; {procedure box}

procedure introscreen;
 begin
  clrscr; mark;
  box(10,3,70,20,1); unmark;
  box(24,6,56,8,1);box(22,5,58,9,1);
  box(26,10,55,16,1);
  GOTOXY(25,7); highvideo;
  write('FIVE pH POINT TITRATION METHOD');normvideo;
  GOTOXY(28,11);
  write('FOR DETERMINATION OF:');
  GOTOXY(28,13);
  write('(1) SHORT-CHAIN FATTY ACIDS');
  GOTOXY(28,15);
  write('(2) H2CO3*ALKALINITY');
  GOTOXY(11,16);
  GOTOXY(12,18);
  write('Copyright: UCT');
  box(10,22,41,24,1);
  GOTOXY(12,23);
  write('Press any letter to continue');
  intro := readkey;
 end;  {procedure introscreen}

procedure initlabel;
 begin
  labels[1]:='pHo (initial pH)................';
  labels[2]:='pH1 (after adding Vx1)..........';
  labels[3]:='pH2 (after adding Vx2)..........';
  labels[4]:='pH3 (after adding Vx3)..........';
  labels[5]:='pH4 (after adding Vx4)..........';
  labels[6]:='   Vx1 (ml).....................';
  labels[7]:='   Vx2 (ml).....................';
  labels[8]:='   Vx3 (ml).....................';
  labels[9]:='   Vx4 (ml).....................';
  labels[10]:='Normality of titrant (mol/l)....';
  labels[11]:='Sample size: undiluted (ml).....';
  labels[12]:='Sample size: diluted (ml).......';
  labels[13]:='Temperature (Celsius)...........';
  labels[14]:='TDS (mg/l)......................';
  labels[15]:='Specific Conductivity (mS/m)....';
  labels[16]:='Inorganic Nitrogen (mgN/l)......';
```

```
  labels[17]:='Inorganic Phosphorus (mgP/l)....';
 end;  {procedure initlabel}

procedure default_values;
 begin
  value[1]:= 7.36;
  value[2]:= 6.75;
  value[3]:= 5.95;
  value[4]:= 5.18;
  value[5]:= 4.29;
  value[6]:= 1.06;
  value[7]:= 3.50;
  value[8]:= 4.84;
  value[9]:= 5.40;
  value[10]:= 0.0728;
  value[11]:= 10;
  value[12]:= 50;
  value[13]:= 21;
  value[14]:= 3300;
  value[15]:= 488;
  value[16]:= 0;
  value[17]:= 0;
 end;  {procedure default_values}

procedure allocate_data;
 begin
  pH0:= value[1];
  pH1:= value[2];
  pH2:= value[3];
  pH3:= value[4];
  pH4:= value[5];
  vx1:= value[6];
  vx2:= value[7];
  vx3:= value[8];
  vx4:= value[9];
  ca:= value[10];
  vsundil:= value[11];
  vsdil:= value[12];
  temp:= value[13];
  TDS:= value[14];
  ionstr:= value[15];
  Nt:= value[16];
  Pt:= value[17];
 end;  {allocate_data}

procedure write_value;
 begin
  GOTOXY(37,2+posy);
   case posy of
          1,2,3,4,5,6,7,8,9: begin
                               writeln(value[posy]:1:2);
                             end;
                         10: begin
                              writeln(value[posy]:1:4);
                             end;
    11,12,13,14,15,16,17,18: begin
                              writeln(value[posy]:1:0);
                             end;
                             else
                             end;
 end;   {procedure write_value}

procedure screen;
 begin
  mark; box(50,2,79,8,1); unmark;
  box(50,10,79,19,1);
  box(50,10,59,19,1);
  box(50,10,79,13,1);
  GOTOXY(52,4);
```

```
  write(' 5 pH POINT TITRATION ');
  GOTOXY(52,6);
  write(' TITRATION INPUT DATA');
  GOTOXY(52,12);
  writeln('  KEY      FUNCTION');
  GOTOXY(52,14);
  writeln('    ',CHR(24),CHR(25),'    SELECT PARAMETER');
  GOTOXY(52,15);
  writeln('<enter>  ERASE VALUE');
  GOTOXY(52,16);
  writeln('<enter>  INSERT NEW VALUE');
  GOTOXY(52,17);
  writeln('   C     CALCULATION');
  GOTOXY(52,18);
  writeln('   Q     QUIT');
 end; {procedure sreeen}

procedure restore;
 begin
  clrscr;
  screen;
  mark; box(1,1,47,22,1); unmark;
   FOR I:= 1 to 17 Do
    begin
     GOTOXY(3,2+I);
     writeln(labels[I]);
     GOTOXY(37,2+I);
      case I of
             1,2,3,4,5,6,7,8,9: begin
                                 writeln(value[I]:1:2);
                                end;
                            10: begin
                                 writeln(value[I]:1:4);
                                end;
      11,12,13,14,15,16,17,18: begin
                                 writeln(value[I]:1:0);
                                end;

      else
      end;
    end;  {FOR}
  GOTOXY(3,2+posy);
  mark;
  write(labels[posy]);
  write_value;
  unmark;
 end;

procedure display;
 begin
  clrscr;
  screen;
  mark; box(1,1,47,21,1); unmark;
  FOR I:= 1 to 17 Do
   begin
    if I=1 then mark;
    GOTOXY(3,2+I);
    writeln(labels[I]);
    GOTOXY(37,2+I);
   case I of
          1,2,3,4,5,6,7,8,9: begin
                              writeln(value[I]:1:2);
                             end;
                         10: begin
                              writeln(value[I]:1:4);
                             end;
    11,12,13,14,15,16,17,18: begin
                              writeln(value[I]:1:0);
                             end;
   else
```

```
   end;
   if I=1 then unmark;
  end;
 posy:=1;
 REPEAT
 repeat move := readkey until (upcase(move) in ['C','Q']) or (ORD(move)
                                            in [72,80,13]);
  case ORD(move) of
   80: begin
        GOTOXY(3,2+posy);
        writeln(labels[posy]);
        write_value;
        posy:=posy+1;
        if posy=18 then posy:=1;
        GOTOXY(3,2+posy);
        mark; writeln(labels[posy]); write_value; unmark;
       end;
   72: begin
        GOTOXY(3,2+posy);
        writeln(labels[posy]);
        write_value;
        posy:=posy-1;
        if posy=0 then posy:=17;
        GOTOXY(3,2+posy);
        mark; writeln(labels[posy]); write_value; unmark;
       end;
   13: begin
        repeat
         GOTOXY(37,2+posy);
         mark;
         writeln('       ');
         GOTOXY(37,2+posy);
         {$I-} read(number) {$I+}
        until ioresult = 0;
        value[posy]:= number;
        GOTOXY(37,2+posy);
        unmark;
        writeln('        ');
        write_value;
        case posy of
          14: begin
               value[posy+1]:= 0.1488*(value[posy]-20);
               if value[posy+1] < 0 then value[posy+1]:= 0;
               GOTOXY(37,posy+3);
               write('        ');
               GOTOXY(37,posy+3);
               write(value[posy+1]:1:0);
              end;
         15: begin
              value[posy-1]:= 6.72*value[posy]+20;
              GOTOXY(37,posy+1);
              write('        ');
              GOTOXY(37,posy+1);
              write(value[posy-1]:1:0);
             end;
        else
        end;
       end;
  else
  end;

  until upcase(move) in ['C','Q'];
  clrscr; writeln('Calculating, please wait');
 end;  {procedure display}


procedure pK ;
 begin
  ktemp:= 273 + temp;
```

```
  if TDS < 20 then TDS:= 21;
  logf1:= logf(mue(TDS,dil),ktemp);
  logf2:= 4*logf1;
  pK1:= -1*(-356.3094 - 0.06091964*ktemp + 21834.37/ktemp + 126.8339
        *log(ktemp) - 1684915/(ktemp*ktemp));
  pK11:= pK1 + logf1;
  pK2:= -1*(-107.8871 - 0.03252849*ktemp + 5151.79/ktemp + 38.92561
        *log(ktemp) - 563713.9/(ktemp*ktemp));
  pK22:= pK2 - logf1 + logf2 ;
  pKa:= 1170.5/ktemp - 3.165 + 0.0134 * ktemp;
  pKaa:= pKa + logf1;
  pKn:= 2835.8/ktemp -0.6322 + 0.00123 * ktemp;
  pKnn:= pKn + logf1 ;
  pKp:= 1979.5/ktemp - 5.3541 + 0.01984 * ktemp;
  pKpp:= pKp - logf1 + logf2 ;
 end;  {procedure pK}

procedure deltapH;
 begin
  pH0:= pH0 + pHcorr;  pH1:= pH1 + pHcorr;  pH2:= pH2 + pHcorr;
  pH3:= pH3 + pHcorr;  pH4:= pH4 + pHcorr;
 end;

procedure atctcalculation;
 begin
  A1:= (vx2-vx1)*ca - MH2O(vx1,vx2,pH1,pH2) - MNH3(pH1,pH2) -
       MHPO4(pH1,pH2) + dHAcalk(pH1,pH2)/dHAcalk(pH3,pH4) *
       (MH2O(vx3,vx4,pH3,pH4) + MNH3(pH3,pH4) + MHPO4(pH3,pH4) -
       (vx4-vx3)*ca);
  B1:= dH2CO3alk(pH1,pH2) - dHAcalk(pH1,pH2)/dHAcalk(pH3,pH4) *
       dH2CO3alk(pH3,pH4);

  A2:= (vx4-vx1)*ca - MH2O(vx1,vx4,pH1,pH4) - MNH3(pH1,pH4) -
       MHPO4(pH1,pH4) + dHAcalk(pH1,pH4)/dHAcalk(pH3,pH4) *
       (MH2O(vx3,vx4,pH3,pH4) + MNH3(pH3,pH4) + MHPO4(pH3,pH4) -
       (vx4-vx3)*ca);
  B2:= dH2CO3alk(pH1,pH4) - dHAcalk(pH1,pH4)/dHAcalk(pH3,pH4) *
       dH2CO3alk(pH3,pH4);

 Ct1:= (A1/B1)/vsdil * 50000 * dil;            {using pH: 1-2; 3-4}
 Ct2:= (A2/B2)/vsdil * 50000 * dil;            {using pH: 1-4; 3-4}
 MCt1:= A1/B1;
 Ctcomp:= Ct1 - Ct2 ;
end;  {procedure atctcalculation}

procedure atct1;
 begin
  dil:= vsdil/vsundil;
  pK; delpH:= 0; pHcorr:=0; atctcalculation; counter:= 0;
  MAt1:= 1/dHAcalk(pH3,pH4) * ((vx4-vx3)*ca-MCt1*dH2CO3alk(pH3,pH4) -
  MNH3(pH3,pH4) - MHPO4(pH3,pH4) - MH2O(vx3,vx4,pH3,pH4));
  At1:= MAt1/vsdil * 60000 * dil;
  CtAtratio:= At1/Ct1;
  if Ctcomp = 0 then x:= 'd';
  if Ctcomp > 0 then x:= 'a';
  if Ctcomp < 0 then x:= 'b';
  if CtAtratio > 0.5 then x:= 'c';
   case x of
     'a': begin
           pHcorr:= -0.01;
            repeat
             delpH:= delpH - 0.01;
             deltapH; counter:= counter + 1;
             atctcalculation;
            until (Ctcomp < 0) or (counter > 19);
          end;
     'b': begin
           pHcorr:= 0.01;
            repeat
```

```
            delpH:= delpH + 0.01;
            deltapH; counter:= counter + 1;
            atctcalculation;
           until (Ctcomp > 0) or (counter > 19);
          end;
     else
     end;

   MAt1:= 1/dHAcalk(pH3,pH4) * ((vx4-vx3)*ca-MCt1*dH2CO3alk(pH3,pH4) -
          MNH3(pH3,pH4) - MHPO4(pH3,pH4) - MH2O(vx3,vx4,pH3,pH4));
   At1:= MAt1/vsdil * 60000 * dil;

   MAt2:= 1/dHAcalk(pH1,pH4) * ((vx4-vx1)*ca-MCt1*dH2CO3alk(pH1,pH4) -
          MNH3(pH1,pH4) - MHPO4(pH1,pH4) - MH2O(vx1,vx4,pH1,pH4));
   At2:= MAt2/vsdil * 60000 * dil;

   MH2CO3alksam:= MCt1 * (perHCO3(pH0) + 2 * perCO3(pH0));
   H2CO3alksam:= MH2CO3alksam / vsdil * dil * 50000 + (tento((pH0-14)) -
                 tento(-pH0)/tento(logf1)) * 50000;
  end;  {procedure atct1}

 procedure output;
  begin
   clrscr;
   mark; box(10,2,60,4,1);unmark;
   box(2,7,73,10,1); box(49,7,73,10,1);
   box(2,11,73,14,1);  box(49,11,73,14,1);
   box(2,15,73,18,1); box(49,15,73,18,1);
   GOTOXY(26,3);
   highvideo; write('OUTPUT DATA '); normvideo;
   GOTOXY(5,8);
   write('H2CO3*alkalinity (undiluted sample)');
   GOTOXY(58,8);
   highvideo; write(H2CO3alksam:3:0); normvideo;
   GOTOXY(5,9);
   write('(mg/l as CaCO3)');
   GOTOXY(5,12);
   write('Short-chain fatty acids (undiluted sample)');
   GOTOXY(58,12);
   if At1 > 0 then
    begin
     highvideo; write(At1:3:0); normvideo;
    end
    else
    begin
     highvideo; write('0 ');normvideo;
    end;
   GOTOXY(5,13);
   write('(mg/l as acetic acid)');
   GOTOXY(5,17);
   write('Systematic pH error');
   GOTOXY(58,17);
   highvideo; write(delpH:1:2); normvideo;
   if counter = 20 then
    begin
     GOTOXY(5,16);
     write('The titration data indicate a systematic');
     GOTOXY(5,17);
     write('pH error > 0.2; Check pH probe calibration');
     GOTOXY(58,17);
     write('      ');
    end;
   if x = 'c' then
    begin
     GOTOXY(5,16);
     write('Correction for systematic pH error');
     GOTOXY(5,17);
     write('is not possible for this titration');
     GOTOXY(60,17);
```

```
      write('      ');
     end;
   end; {procedure output}

  procedure exitbox;
   begin
    repeat;
     box(5,20,60,24,1); box(49,20,60,22,1);
     box(5,22,60,24,1); box(49,22,60,24,1);
     GOTOXY(7,21);
     write('Do you wish to do a further calculation ?');
     GOTOXY(53,21);
     highvideo; write('Y'); normvideo; write('/'); highvideo; write ('N');
     normvideo;
     GOTOXY(7,23);
     write('Do you wish to quit the program ?');
     GOTOXY(54,23);
     highvideo; write('Q'); normvideo;
     again:=readkey; again:=upcase(again);
    until again in ['Y','N','Q'];
    clrscr;
   end;  {procedure exitbox}



   begin                  { of main program}

     introscreen;
     again:= 'Y';
     default_values;

     repeat; clrscr; initlabel; display;
      if upcase(move) in ['C'] then
       begin
        allocate_data; atct1; output; exitbox;
       end
       else
       begin
        clrscr; again:= 'N';
       end;
     until again in ['N','Q'];

 end.   {of main program}

{PROGRAM END.}
```

APPENDIX X

INSTRUCTIONS TO RUN THE EXECUTABLE FILES OF 4 AND 5 pH POINT TITRATION PERSONAL COMPUTER PROGRAMS

Starting program execution

Floppy disk system:

- Boot up the computer from a DOS disk in drive A:
- Remove the DOS disk from drive A: and replace with the disk attached to the inside of the thesis cover
- To initiate the execution of the program for the ***4 pH point titration***, at the DOS prompt A:> the user must type

  4ppt.exe

  and then press the carriage return key (<RETURN>, <C/R>, <ENTER> key on different keyboards).
- To initiate the execution of the program for the ***5 pH point titration***, at the DOS prompt A:> the user must type

  5ppt.exe

  and then press the carriage return key.
- The program will be loaded into memory from disk. Depending on the choice of programs one of the title pages shown in Table X.1 and Table X.2 respectively, will appear on the screen:

Table X.1: Title page for the 4 pH point titration method:

```
FOUR pH POINT TITRATION METHOD

FOR DETERMINATION OF:

(1) H2CO3*ALKALINITY
(2) TOTAL CARBONATE SPECIES CONCENTRATION (Ct)

Copyright: UCT

Press any letter to continue
```

Table X.2: Title page for the 5 pH point titration method:

```
FIVE pH POINT TITRATION METHOD

FOR DETERMINATION OF:

(1) SHORT-CHAIN FATTY ACIDS

(2) H2CO3*ALKALINITY

Copyright: UCT

Press any letter to continue
```

Hard disk system:

- Boot up the system from the hard disk
- To initiate the execution of the program for the ***4 pH point titration***, at the DOS prompt C:> the user must type

  a:4ppt.exe

  and then press the carriage return key.
- To initiate the execution of the program for the ***5 pH point titration***, at the DOS prompt C:> the user must type

  a:5ppt.exe

  and then press the carriage return key.
- The program will be loaded into memory from disk. Depending on the choice of programs one of the above title pages will appear on the screen.

Data input by the user

With the title page present on the screen a data input table will appear on the screen if any letter on the keyboard is being pressed. For the 4 pH point titration program the data input table shown in Table X.3 will appear on the screen. For the 5 pH point titration program the input data table shown in Table X.4 will appear.

Table X.3: Data input table for 4 pH point titration method.

pHo (initial pH)................	8.00
pH1 (after adding Vx1)..........	6.73
pH2 (after adding Vx2)..........	5.93
pH3 (after adding Vx3)..........	5.12
Vx1 (ml)....................	0.40
Vx2 (ml)....................	1.00
Vx3 (ml)....................	1.30
Normality of titrant (mols/l)...	0.0728
Sample size (undiluted) (ml)....	50
Sample size (diluted) (ml)......	50
Temperature (Celsius)..........	21
TDS (mg/l)......................	160
Ionic strength (mS/m)...........	16
Inorganic Nitrogen (mgN/l)......	0
Inorganic Phosphorus (mgP/l)....	0

4 pH POINT TITRATION

TITRATION INPUT DATA

KEY	FUNCTION
↑↓	SELECT PARAMETER
<enter>	ERASE VALUE
<enter>	INSERT NEW VALUE
C	CALCULATION
Q	QUIT

Table X.4: Data input table for 5 pH point titration method.

pHo (initial pH)...............	7.36
pH1 (after adding Vx1)..........	6.75
pH2 (after adding Vx2)..........	5.95
pH3 (after adding Vx3)..........	5.18
pH4 (after adding Vx4)..........	4.29
Vx1 (ml)....................	1.06
Vx2 (ml)....................	3.50
Vx3 (ml)....................	4.84
Vx4 (ml)....................	5.40
Normality of titrant (mol/l)....	0.0728
Sample size: undiluted (ml).....	10
Sample size: diluted (ml).......	50
Temperature (Celsius)...........	21
TDS (mg/l)......................	3300
Specific Conductivity (mS/m)....	488
Inorganic Nitrogen (mgN/l)......	0
Inorganic Phosphorus (mgP/l)....	0

5 pH POINT TITRATION

TITRATION INPUT DATA

KEY	FUNCTION
↑↓	SELECT PARAMETER
<enter>	ERASE VALUE
<enter>	INSERT NEW VALUE
C	CALCULATION
Q	QUIT

Both data input tables contain default values for the different input parameters; the first parameter and default value (pH_0) will be highlighted. In order to change the default value (for the highlighted parameter) the user must (1) press the carriage return key to erase the existing value, (2) type in the new value, and (3) press the carriage return key again. If the default or the entered value is acceptable the following or previous parameter is selected with the arrow keys: ↓ or ↑ respectively. In order to invoke the calculation part of the program the user must type the letter C. After termination of the calculations the output data table (table of result) will appear on the screen.

For the 4 pH point titration program two different output tables may be obtained depending on the magnitude of the systematic pH error: if the systematic pH error is smaller than 0,2 the output table shown in Table X.5 will appear. If the systematic pH error is greater than 0,2 the output table shown in Table X.6 will appear.

<u>Table 5:</u> Output data table for 4 pH point titration program with a systematic pH error smaller than 0,2.

TABLE OF RESULTS	
H2CO3*alkalinity (undiluted sample) (mg/l as CaCO3)	100
Total carbonate species (undiluted sample) (mg/l as CaCO3)	102
Estimate of systematic pH error	-0.03
Do you wish to do a further calculation ?	Y/N
Do you wish to quit the program ?	Q

<u>Table 6:</u> Output data table for 4 pH point titration program with a systematic pH error greater than 0,2.

TABLE OF RESULTS	
H2CO3*alkalinity (undiluted sample) (mg/l as CaCO3)	
Total carbonate species (undiluted sample) (mg/l as CaCO3)	
The titration data indicate a systematic pH error > 0,2; please check pH probe calibration	
Do you wish to do a further calculation ?	Y/N
Do you wish to quit the program ?	Q

Similarly, for the 5 pH point titration program two different output tables will be obtained: if the systematic pH error is smaller than 0,2 the output table shown in Table X.7 will appear. If the systematic pH error is greater than 0,2 the output table shown in Table X.8 will appear.

<u>Table 7:</u> Output data table for 5 pH point titration program with a systematic pH error smaller than 0,2.

OUTPUT DATA	
H2CO3*alkalinity (undiluted sample) (mg/l as CaCO3)	1864
Short-chain fatty acids (undiluted sample) (mg/l as acetic acid)	196
Systematic pH error	-0.03
Do you wish to do a further calculation ?	Y/N
Do you wish to quit the program ?	Q

Table 8: Output data table for 5 pH point titration program with a systematic pH error greater than 0,2.

OUTPUT DATA

H2CO3*alkalinity (undiluted sample) (mg/l as CaCO3)	
Short-chain fatty acids (undiluted sample) (mg/l as acetic acid)	
The titration data indicate a systematic pH error > 0.2; Check pH probe calibration	

Do you wish to do a further calculation ?	Y/N
Do you wish to quit the program ?	Q

With the option keys (Y/N) or (Q) which are displayed near the bottom of the output table, the user may decide to either quit the program or repeat the calculations for a different set of input values.

V3 RAADGEWENDE INGENIEURS (Edms) Bpk

PROJEK			Verw.
Worcester Sewage Works			KS040518
Normalization of acid for titration			Bladsynr. 1 of 1
Tek. Verw.	Bereken deur C. Mellish	Nagegaan deur	Datum 1998.3.19

V_x	pH_x	$F_{x1} = 10^{-pH_x}(V_s + V_x)$
14,62	3,91	0,0079
14,64	3,80	0,0102
14,66	3,69	0,0132
14,68	3,58	0,0170
14,69	3,52	0,0195

Use Linear Regression to calculate $V_{xo} = 14{,}58$

Clear Σ registers

0,0079 <Enter> 14,62 $\Sigma+$ → 1.00
0,0102 <Enter> 14,64 $\Sigma+$ → 2.00
0,0132 <Enter> 14,66 $\Sigma+$ → 3.00
0,0170 <Enter> 14,68 $\Sigma+$ → 4.00
0,0195 <Enter> 14,69 $\Sigma+$ → 5.00

Place a value of 0 in x register and select $\hat{x}$ under Linear Regression to obtain value for $V_{xo} = 14{,}58$